

Implementation of Centralized SSH Monitoring and Detection in Splunk Cloud

BY

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EXECUTIVE SUMMARY

DataFortis Technologies, a cybersecurity and cloud infrastructure provider, noticed an increase in SSH-related security events across its cloud environment. These events included multiple failed login attempts, unauthorized access attempts, and signs of brute-force activity.

The project was initiated to improve how SSH activity is monitored and detected across the DataFortis cloud infrastructure. Prior to implementation, the SOC lacked a unified view of SSH events, limiting the ability to quickly identify suspicious behaviour, correlate activity across systems, and respond effectively to emerging threats.

As part of the project, centralized SSH log collection was implemented along with role-based access controls for SOC analysts and engineering teams. SOC-focused dashboards were created to provide clear visibility of SSH activity, and automated alerts were configured to notify the SOC of suspicious login attempts and abnormal patterns.

Overall, the project improved DataFortis Technologies security monitoring capabilities by allowing the SOC to detect, investigate, and escalate SSH-related threats in near real time. This has helped strengthen the security of the cloud infrastructure, improve incident response efficiency, and support ongoing compliance and security best practices.

OBJECTIVE

- To improve detection of anomalous SSH activity, including brute-force attempts, attempts from unusual IP & attempts from unusual geolocations in real time.
- To create and provide role-specific SOC dashboards for analysts, SOC leads, and security engineers to support operational monitoring and risk visibility.
- To Implement automated alerts and standardized runbooks for consistent SSH threat triage and escalation.
- To Strengthen Splunk platform security through role-based access aligned with SOC tiers and their required permission.

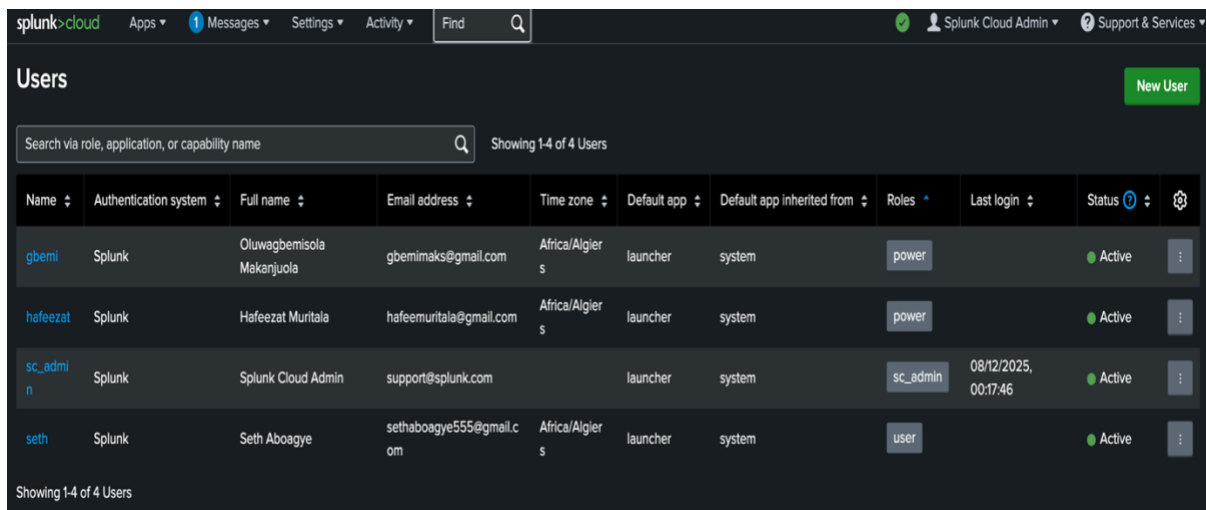
SPLUNK CLOUD ACCOUNT CREATION AND USER CONFIGURATION

This report outlines the configuration settings applied during the setup of new users in the Splunk Cloud environment. It includes user identity details, assigned roles, and system-wide configuration policies to ensure secure and role-appropriate access.

Configuration Standards Applied

1. **Time Zone:** All users are configured with the time zone set to WAT (West Africa Time), represented in Splunk as Africa/Algiers.
2. **Default App:** The default application for all users is set to launcher (home), inherited from the system.
3. **Password Policy:** All user accounts are configured to require password change upon first login to enforce credential hygiene.
4. **Role Assignment:** The roles are assigned based on user responsibilities within the Splunk environment.

User Details and Role Assignments



The screenshot shows the 'Users' management page in Splunk Cloud. At the top, there's a navigation bar with 'splunk>cloud', 'Apps', 'Messages', 'Settings', 'Activity', and a search bar. Below this, the 'Users' section has a 'New User' button and a search bar. The main content is a table with 10 columns: Name, Authentication system, Full name, Email address, Time zone, Default app, Default app inherited from, Roles, Last login, and Status. There are 4 users listed, all with 'Active' status. The roles are 'power', 'power', 'sc_admin', and 'user'.

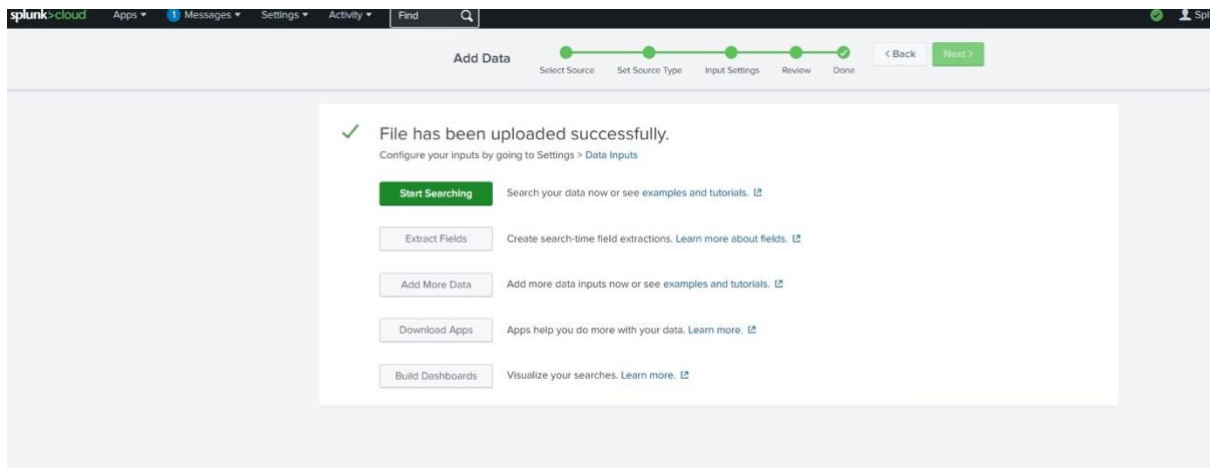
Name	Authentication system	Full name	Email address	Time zone	Default app	Default app inherited from	Roles	Last login	Status
gbemi	Splunk	Oluwagbemisola Mekanjuola	gbemimaks@gmail.com	Africa/Algiers	launcher	system	power		Active
hafeezat	Splunk	Hafeezat Muritala	hafeemuritala@gmail.com	Africa/Algiers	launcher	system	power		Active
sc_admin	Splunk	Splunk Cloud Admin	support@splunk.com		launcher	system	sc_admin	08/12/2025, 00:17:46	Active
seth	Splunk	Seth Aboagye	sethaboagye555@gmail.com	Africa/Algiers	launcher	system	user		Active

Role Descriptions

1. **power:** It is assigned to users with elevated privileges for data access and dashboard creation.
2. **user:** It is assigned to standard users with limited access for data viewing.
3. **sc_admin:** It is reserved for administrative tasks including system configuration and user management.

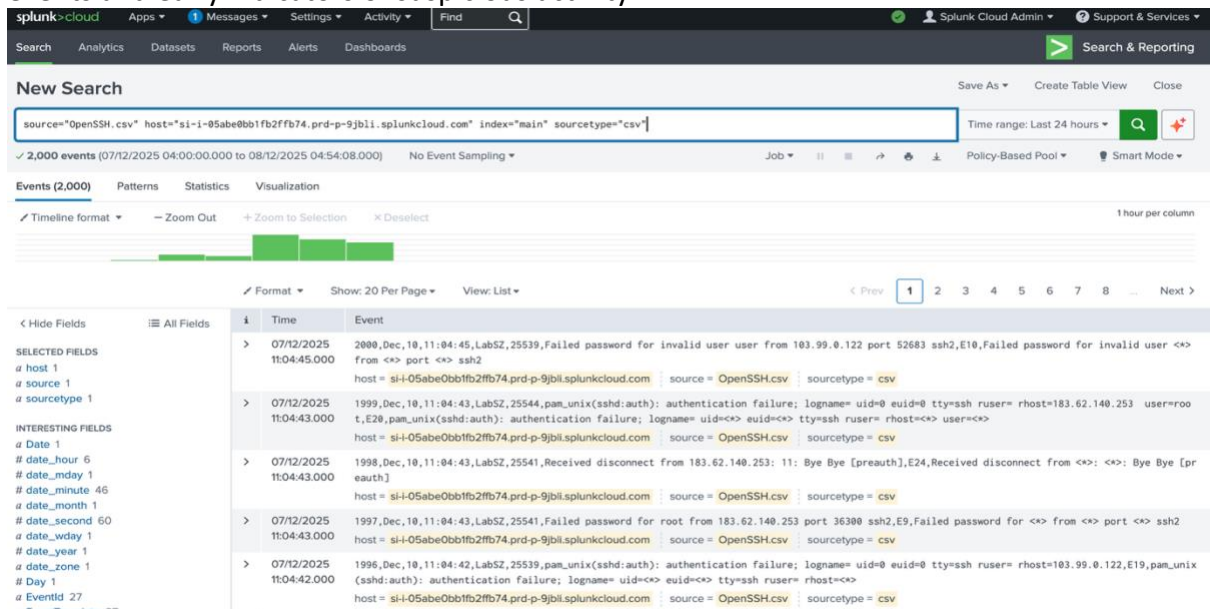
DATA INGESTION INTO SPLUNK CLOUD

The OpenSSH.csv log file has been successfully uploaded into the Splunk Cloud environment using the Add Data workflow. All steps in the ingestion pipeline—Select Source, Set Source Type, Input Settings, Review, and Done were completed without error.



Log File Analysis – Splunk Cloud

This report documents the initial observations made from the uploaded log data in Splunk Cloud before detailed analysis commenced. The review focuses on the volume of ingested events and early indicators of suspicious activity.



Event Summary

The Total Number of Events Indexed: 2,000 events

- This count reflects the volume of log entries successfully ingested and indexed by Splunk Cloud from the uploaded source.

Suspicious Notifications Identified

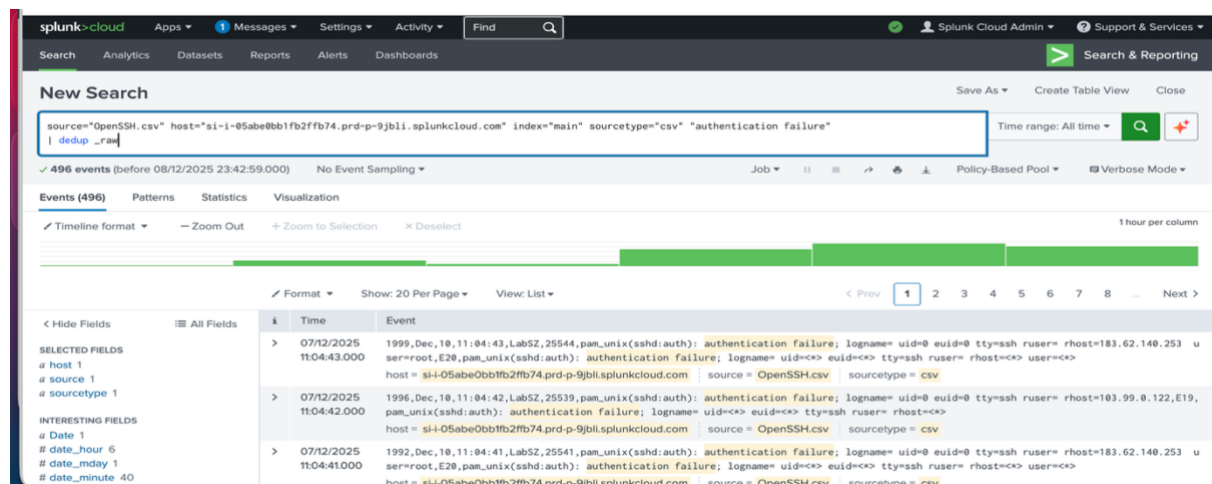
Before full analysis began, the following suspicious log entries were observed:

1. **Repeated Authentication Failures:** Multiple entries were flagged with failed login attempts, suggesting potential brute-force or unauthorized access attempts.
2. **Unusual SSH Activity:** Log lines referencing sshd with error codes and repeated connection attempts from similar IP addresses.
3. **Alert Keywords Detected:** Terms such as “Failed password”, “Disconnected from” and “Invalid user” appeared frequently, indicating possible reconnaissance or credential stuffing behaviour.

Investigation was carried out on the suspicious “Authentication Failure”

An SPL query search was used to detect potential unauthorized access attempts and assess whether any suspicious or malicious activity was present.

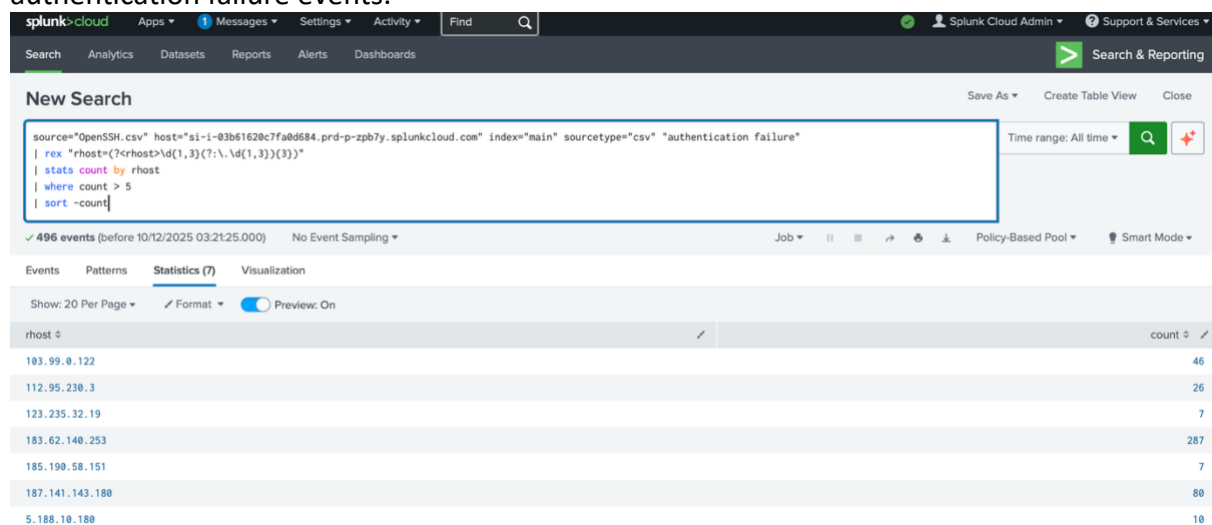
The search returned 496 events, all categorized as “authentication failure” within the SSH authentication logs.



Number of Authentication failures Events(496 events)

Further Investigation into the Authentication attempts from remote host(rhost):

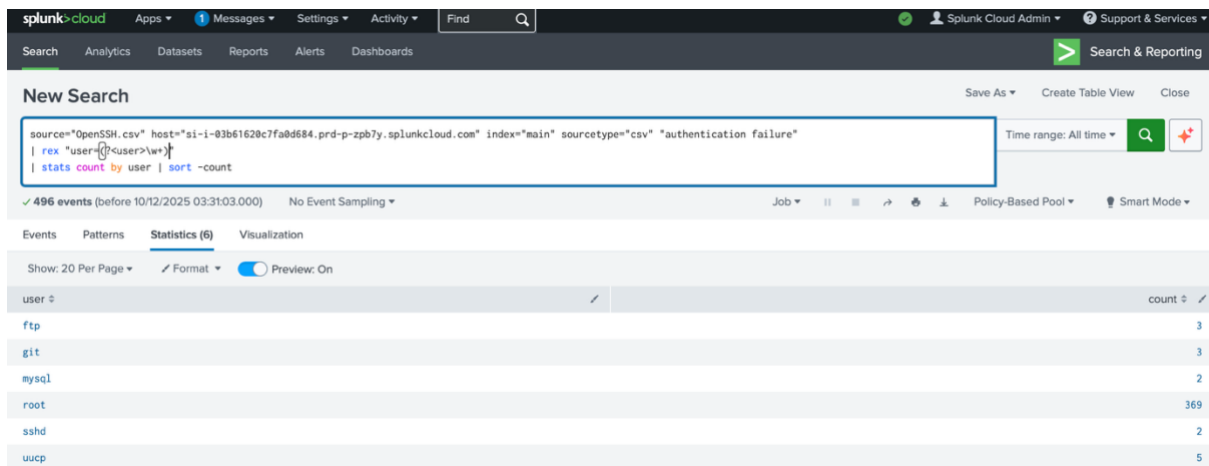
The Splunk SPL was queried to identify remote IP addresses (rhost) with more than five authentication failure events.



Number of Authentication failures by each remote host(rhost).

Investigation into Authentication Failure by Users

The Splunk SPL query search was further queried to identify user accounts with authentication failure events. The goal is to detect which user are most frequently targeted or involved in failed login attempts, which may indicate brute-force attacks, misconfigurations, or unauthorized access attempts.



New Search

source='OpenSSH.csv' host='si-i-03b61628c7fa0d684.prd-p-zpb7y.splunkcloud.com' index='main' sourcetype='csv' 'authentication failure'
 | rex 'user=(?<user>\w*)'
 | stats count by user | sort -count

✓ 496 events (before 10/12/2025 03:31:03.000) No Event Sampling

Events Patterns **Statistics (6)** Visualization

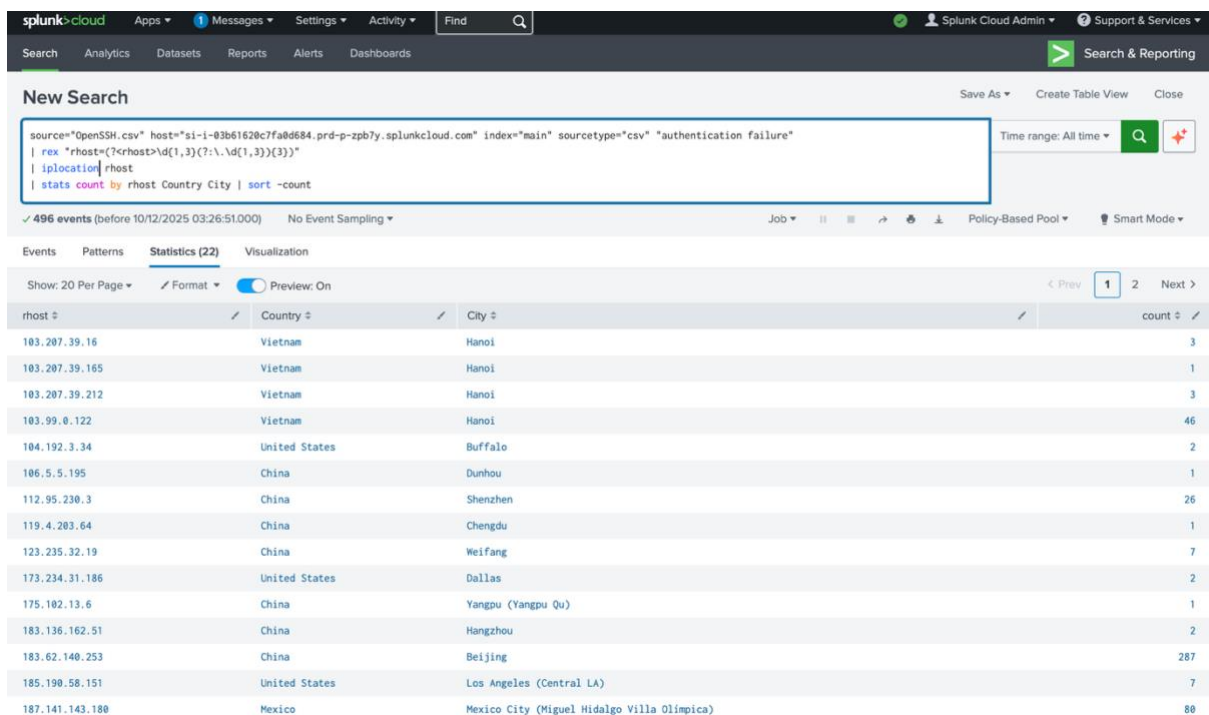
Show: 20 Per Page Format Preview: On

user	count
ftp	3
git	3
mysql	2
root	369
sshd	2
uucp	5

The above screenshot indicates that the (root) user was the most targeted.

Investigation into Authentication Failure from Unusual Geographic region

Further analysis was carried out on authentication failure logs to identify unusual login attempts originating from unexpected geographic locations.



New Search

source='OpenSSH.csv' host='si-i-03b61628c7fa0d684.prd-p-zpb7y.splunkcloud.com' index='main' sourcetype='csv' 'authentication failure'
 | rex 'rhost=(?<rhost>\d{1,3}(?:\.\d{1,3}){3})'
 | iplocation rhost
 | stats count by rhost Country City | sort -count

✓ 496 events (before 10/12/2025 03:26:51.000) No Event Sampling

Events Patterns **Statistics (22)** Visualization

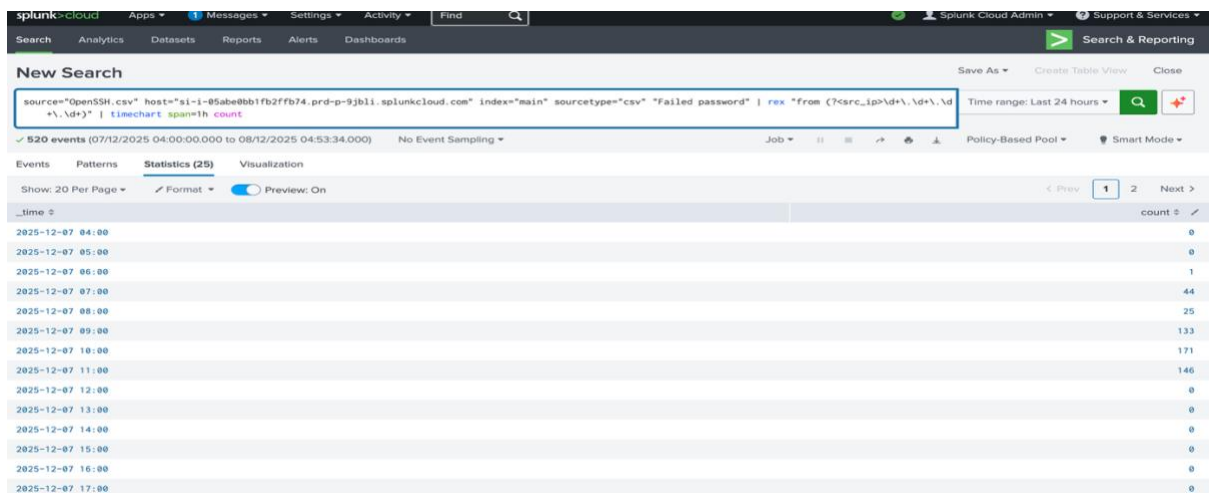
Show: 20 Per Page Format Preview: On

rhost	Country	City	count
103.207.39.16	Vietnam	Hanoi	3
103.207.39.165	Vietnam	Hanoi	1
103.207.39.212	Vietnam	Hanoi	3
103.99.0.122	Vietnam	Hanoi	46
104.192.3.34	United States	Buffalo	2
106.5.5.195	China	Dunhou	1
112.95.230.3	China	Shenzhen	26
119.4.203.64	China	Chengdu	1
123.235.32.19	China	Weifang	7
173.234.31.186	United States	Dallas	2
175.102.13.6	China	Yangpu (Yangpu Qu)	1
183.136.162.51	China	Hangzhou	2
183.62.140.253	China	Beijing	287
185.190.58.151	United States	Los Angeles (Central LA)	7
187.141.143.180	Mexico	Mexico City (Miguel Hidalgo Villa Olimpica)	80

Number of Authentication failures originating from different Geographic region

Investigation into Failed Password Trend & Unusual IP addresses attempt Events.

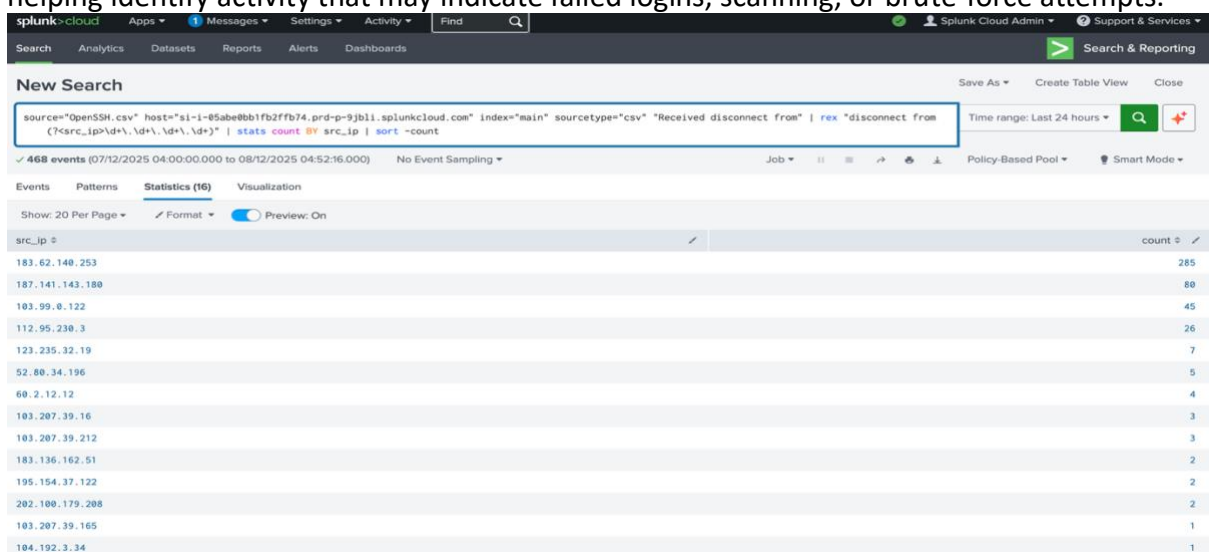
The investigation focused on detecting failed login attempts and identifying unusual IP addresses that may indicate malicious activity.



A significant spike in failed login attempts occurred between 09:00 and 11:00 on December 5, 2025, peaking at 171 events at 10:00. Such elevated levels of authentication failures within consecutive hours are consistent with: **Brute-force attacks, Unauthorized scanning activity & Credential-stuffing attempts.**

Unusual IP Address Activity

The investigation reviewed disconnection events from IP addresses attempting to connect to the infrastructure. A Splunk query using rex was used to extract the source IP addresses, helping identify activity that may indicate failed logins, scanning, or brute-force attempts.

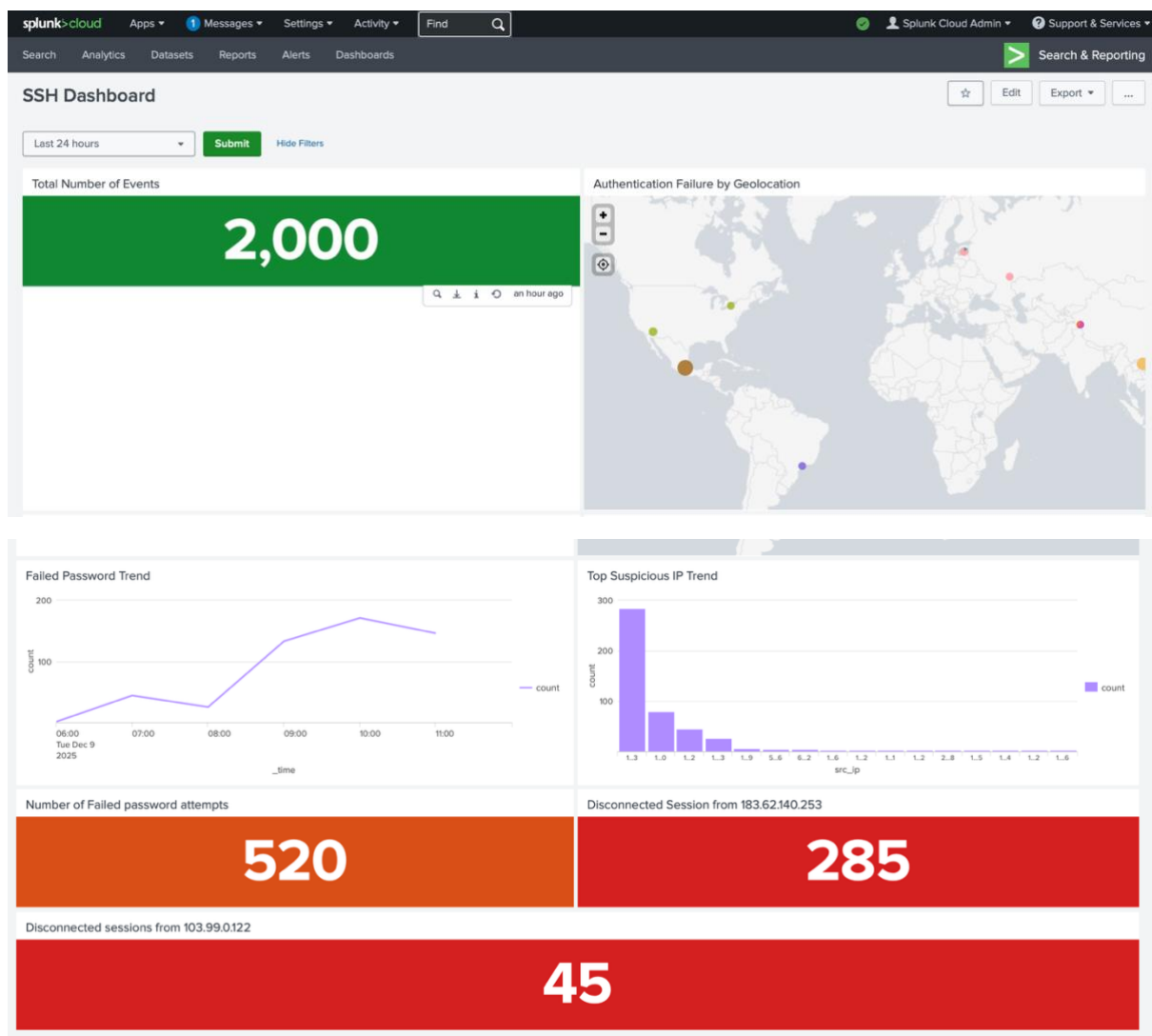


- 183.62.140.253 stands out with 285 disconnects, suggesting persistent and possibly automated access attempts.
- 187.141.143.180 and 103.99.0.122 also show elevated activity, which may be indicative of coordinated probing or distributed brute-force behaviour.
- Threat Intelligence Correlation: These IPs were checked against known malicious IP database (AbuseIPDB), and they were all tagged malicious for brute force and DDOS attacks.

DASHBOARD CREATION (SSH DASHBOARD)

The Splunk dashboard provides centralized visibility by transforming raw logs into actionable insights, enabling cross-system correlation, real-time monitoring, improved incident response, and support for reporting and compliance requirements.

A dashboard (SSH dashboard) was created using logs from the previous investigation to analyse SSH authentication activity. This dashboard consolidates key security metrics, including total SSH event counts, authentication failures from unusual geolocations, failed password trends, suspicious IP activity, and disconnection statistics associated with malicious source IPs.



Dashboard Panels and Insights

Total number of events (2,000): The dashboard tells us the total number of events which includes authentication attempts, disconnection events, and failure attempts logs.

Splunk dashboard Query:

```
source="OpenSSH.csv" host="si-i-03b61620c7fa0d684.prd-p-zpb7y.splunkcloud.com"
sourcetype="csv" | stats count
```


Authentication Failure from Unusual Geolocation: This highlights the geographic markers indicating where several authentication failures occurred. This helps attribute suspicious activity to specific regions, supporting geo-based threat intelligence.

Splunk dashboard Query:

```
source="OpenSSH.csv" host="si-i-03b61620c7fa0d684.prd-p-zpb7y.splunkcloud.com"
index="main" sourcetype="csv" "authentication failure" | rex
"rhost=(?<rhost>\d+\.\d+\.\d+\.\d+)"
| iplocation rhost
| geostats count by Country
```

Failed Password Trend: This visualizes the temporal pattern of failed SSH login attempts. A noticeable spike around 10:00 suggests a concentrated brute-force attempt during that hour.

Splunk dashboard Query:

```
source="OpenSSH.csv" host="si-i-03b61620c7fa0d684.prd-p-zpb7y.splunkcloud.com"
index="main" sourcetype="csv" "Failed password" | rex "from (?<src_ip>\d+\.\d+\.\d+\.\d+)"
| timechart span=1h count
```

Top Suspicious IP Trend: The chart highlights the source IPs responsible for high volumes of suspicious login activity.

Splunk dashboard Query:

```
source="OpenSSH.csv" host="si-i-03b61620c7fa0d684.prd-p-zpb7y.splunkcloud.com"
index="main" sourcetype="csv" "Received disconnect from" | rex "disconnect from"
(?<src_ip>\d+\.\d+\.\d+\.\d+)" | stats count BY src_ip | sort -count
```

Number of Failed Password Attempts – 520: This visualizes the total number of failed password attempts detected in the logs. This metric helps assess the scale of brute-force activity and supports alert prioritization.

Splunk dashboard Query:

```
source="OpenSSH.csv" host="si-i-03b61620c7fa0d684.prd-p-zpb7y.splunkcloud.com"
index="main" sourcetype="csv" "failed password" | stats count
```

Disconnected Session from 183.62.140.253 (285 events) & 103.99.0.122 (45 events): This indicates a high number of disconnected sessions from these IPs, suggesting possible session hijacking, forced terminations, or abnormal behaviour from this source.

Splunk dashboard Query:

```
source="OpenSSH.csv" host="si-i-03b61620c7fa0d684.prd-p-zpb7y.splunkcloud.com"
index="main" sourcetype="csv" "Received disconnect from 183.62.140.253" | stats count
```

AND

```
source="OpenSSH.csv" host="si-i-03b61620c7fa0d684.prd-p-zpb7y.splunkcloud.com"  
index="main" sourcetype="csv" "Received disconnect from 103.99.0.122" | stats count
```

REFINING ALERT CREATION IN SPLUNK CLOUD

This report outlines the creation and configuration of an automated alert in Splunk Cloud to monitor SSH disconnection events originating from the previously identified suspicious IP address 183.62.140.253. The alert was configured to run on a scheduled interval and is triggered when the volume of SSH disconnection events exceeds a predefined threshold. The purpose of this alert is to provide early visibility into potentially unauthorized access attempts or other security-related anomalies associated with this external host, enabling the SOC to quickly investigate and respond to suspicious SSH activity.

Alert Query:

```
source="OpenSSH.csv" host="si-i-03b61620c7fa0d684.prd-p-zpb7y.splunkcloud.com"  
index="main" sourcetype="csv" "Received disconnect from 183.62.140.253"
```

The screenshot shows the 'Save As Alert' configuration window in Splunk Cloud. The background displays a search results page with the query `source="OpenSSH.csv" host="si-i-03b61620c7fa0d684.prd-p-zpb7y.splunkcloud.com" index="main" sourcetype="csv" "Received disconnect from 183.62.140.253"` and 285 events. The 'Save As Alert' dialog is open, showing the following configuration:

- Title:** Disconnect from 183.62.140.253 Alert
- Description:** Disconnect from 183.62.140.253 Alert
- Permissions:** Private (Selected), Shared in App
- Alert type:** Scheduled
- Run every day:** Run every day
- At:** 8:00
- Expires:** 24 hour(s)
- Trigger Conditions:**
 - Trigger alert when: Number of Results
 - is greater than: 2
 - Trigger: Once (Selected), For each result
 - Throttle: ☒
 - Suppress triggering for: 60 second(s)
- Trigger Actions:** (Empty)

Buttons: Cancel, Save

FIELD EXTRACTION CONFIGURATION

This report outlines the procedure and outcome of creating a custom field extraction in Splunk to isolate and analyse source IP addresses captured within the OpenSSH event. The extraction process was performed using Splunk's built-in field extractor and delimiter-based parsing capability. The resulting field, *src_ip* enables visibility into all originating IP addresses present in the dataset and supports further security analysis.

Extract Fields Select Method Rename Fields Save < Back Next >

Rename Fields

Select a delimiter. In the table that appears, rename fields by clicking on field names or values. [Learn more](#)

Delimiter

Space Comma Tab Pipe Other

field1 1940,Dec,10,11:04:23,LabSZ,25508,Failed password for root from 183.62.140.253 port 60554 ssh2,E9,Failed password for <*> from <*> port <*> ssh2

field2 password field3 for field4 root field5 from field6 183.62.140.253 field7 port field8 60554 field9 ssh2,E9,Failed field10 password field11 for field12 <*> field13 from field14 <*> field15 port

Field Name src_ip Rename Field

Preview (17 fields)

Events field1 field2 field3 field4 field5 field6 field7 field8 field9 field10 field11 field12 field13 field14 field15 field16

✓ 1,000 events (12/09/2025 00:00:00.000 to 11/12/2025 03:38:19.000) 20 per page < Prev 1 2 3 4 5 6 7 8 ... Next >

Extract Fields Select Method Rename Fields Save < Back Finish >

Save

Name the extraction and set permissions.

Extractions Name **REPORT:** src_ip

Owner sc_admin

App search

Permissions Owner App All apps

Source type csv

Sample event 1940,Dec,10,11:04:23,LabSZ,25508,Failed password for root from 183.62.140.253 port 60554 ssh2,Failed password for <*> from <*> port <*> ssh2

Fields field1,field2,field3,field4,field5,src_ip,field7,field8,field9,field10,field11,field12,field13,field14,field15,field16,field17

Delimiter space

```
# date_minute 40
a date_month 1
# date_second 60
a date_wday 1
# date_year 1
a date_zone 1
# Day 1
a EventId 27
a EventTemplate 27
a extracted_Host 1
a index 1
# linecount 1
# LinId 100+
# Pid 100+
a punct 32
a splunk_server 1
a src_ip 22
a Status 100+
# timeendpos 5
a Timestamp 100+
# timestartpos 4
```

+ Extract New Fields

src_ip 22 Values, 24.95% of events Selected Yes No

Reports

Top values Top values by time Rare values

Events with this field

Top 10 Values	Count	%
183.62.140.253	287	57.515%
187.141.143.180	80	16.032%
103.99.0.122	46	9.218%
112.95.230.3	26	5.21%
5.188.10.180	12	2.405%
185.190.58.151	10	2.004%
123.235.32.19	7	1.403%
60.2.12.12	5	1.002%
103.207.39.16	3	0.601%
103.207.39.212	3	0.601%

After the src_ip field was created, Splunk populated it across all matching events in the dataset.

CONCLUSION

This report describes how centralized SSH monitoring was successfully set up in Splunk Cloud and how SSH security events were investigated. By collecting and analysing OpenSSH logs, the SOC gained clear visibility into all SSH login activity across the cloud environment.

The investigation found many failed SSH login attempts, with repeated attacks targeting privileged accounts such as the root user. Many of these attempts came from unusual geographic locations, and a short time window showed a sharp increase in failed logins, which is typical of automated brute-force attacks. Several external IP addresses were responsible for repeated connection and disconnection attempts, and threat intelligence checks confirmed these IPs are known to be malicious.

No successful SSH logins or system compromise were identified during the investigation. However, the activity observed shows ongoing attempts to gain unauthorized access. The creation of dashboards, field extractions, and automated alerts has improved the SOC's ability to quickly detect, investigate, and respond to SSH-related threats in real time.

Overall, this project improved DataFortis Technologies security monitoring and incident response capabilities and provides a strong foundation for continued SSH threat detection and analysis.

RECOMMENDATION

Based on the findings of this investigation and SOC best practices, the following recommendations are proposed to further reduce SSH attack surface and enhance security controls:

- **Implement Network-Level Access Controls:** We should apply IP allow-listing or firewall rules to restrict SSH access to only trusted management networks and known administrative IP ranges.
- **Integrate Threat Intelligence Feeds:** We could automate the enrichment of SSH events with external threat intelligence sources to enable faster identification and prioritization of known malicious IP addresses.
- **Enable Account Lockout and Rate Limiting:** We can configure SSH rate limiting or temporary account lockouts after multiple failed logins attempts to mitigate brute-force activity.
- **Regular Dashboard and Alert Review:** We should ensure to review dashboards, alert thresholds, and detection logic to ensure continued effectiveness as threat patterns evolve.
- **Develop and Maintain SSH Incident Runbooks:** We should formalize investigation and response procedures for SSH-related alerts to ensure consistent triage, escalation, and remediation by SOC analysts.

